Portable Microplastic Survey Boat – V2

Critical Component Procurement, Comparisons, Payload and Mission Feasibility

Year 3 Engineering Group Project — preliminary systems design

2 November 2026 — Version 2.0

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Abstract

This Version 2 report retains the Version 1 engineering calculations and adds university-approved supplier listings, traceable MYR prices, explicit quotation gaps and a costed preliminary purchase package. It translates

the requirement to produce a spatial microplastic concentration heat map across a 20 m × 20 m water area within about two hours into a first-pass hull, propulsion, autonomy, pump, communications, and energy architecture. Manufacturer-rated values are cited separately from illustrative engineering assumptions. It does *not* establish an actual motor payload rating, boat top speed, sensor selectivity or two-hour field performance. The companion methodology review contains the scientific measurement screening.

1 Version 2 procurement scope and evidence rules

Procurement rule. The approved/registered supplier list was provided by the project team. A website listing is *not* itself proof of the university's purchase-order status: the department must confirm the current account, delivery and quote requirements. Registered suppliers have no minimum purchase value according to the provided purchase instructions. Non-registered buying is a fallback *only* when a registered supplier cannot fulfil the item, and requires prior Procurement Department coordination with Radha.Sivadasan@nottingham.edu.my.

Price evidence dated 2 November 2026. All amounts in the comparison tables are single-unit **listed MYR prices observed on accessible Malaysian supplier webpages**, not a binding quotation. They may change, and tax, delivery, quantity tiers and compatibility accessories are not included unless a listing explicitly says otherwise. Each compared product has a directly clickable **listing** link, and each item needing a supplier reply is marked **RFQ** (request for quotation). **No unauthorised third-party price is substituted for an approved supplier quote.**

Table 1: Selected approved electronics suppliers and website access. Supplier names and restrictions come from the university project purchase instructions.

Supplier	Verified Malaysian storefront / contact	V2 role
Cytron	my.cytron.io	Itemised public prices
GI Electronic	gie.com.my	Camera price comparison
Element14	my.element14.com	RFQ / alternates
RS	my.rs-online.com	RFQ / protected electricals
DigiKey	digikey.my	RFQ / traceable parts
Autobotic, Mybotic	University-listed vendors; website/product listing not verified here	Direct RFQ
Vanguard KL	Email procurement enquiry	Direct RFQ

Other registered procurement routes: for hull manufacture and waterproof fittings, approach YLT Hardware (email), Grandtech Engineering (email), Multi Impact (email) and LCG Scientific (email). The provided list restricts Dynamics Industrial (email) to *electrical only*. JL Brilliant Resources (email) and TX Moon (email) are the registered *online-items* routes. Their product-specific prices are **RFQ**, not zero.

2 Mission requirements, operating modes and acceptance criteria

A 400 m^2 water survey must yield georeferenced observations and a rendered 2D concentration or explicitly provisional particle proxy map inside 120 minutes. Track A measures aboard a moving or stopped boat; Track B collects known-volume labelled water at 16 illustrative grid locations for portable shoreline analysis. Remote operator retains positive control and a safe manual retrieval plan. The shoreline/boat route, local current, obstacle map, GNSS performance, wind and water depth are not specified and require field reconnaissance.

Table 2: Proposed acceptance gates; numerical criteria are design targets, not published hardware performance.

Dimension	Acceptance demonstration	Consequence if failed
Geospatial output	20×20 m map, original locations, timestamps, size limits, data-quality legend	Map fails stakeholder purpose
Timeliness	First deployment to completed map within ≤ 120 min	Reduce stations or improve throughput
Measurement credibility	Controlled known-particle trials, river interferents, measured sample volume, blanks	Label "suspected MP-like particles" or calibrated proxy
Navigation	Confirm track spacing and positional error; manual RC override	No autonomous close-range operation
Flotation	Float full wet load with reserve buoyancy and stable freeboard	Increase hull displaced volume
Energy	120 min at *measured* average load, BMS/fuse adequate for transients	More battery, lower draw or mission redesign

3 Architecture and critical component list

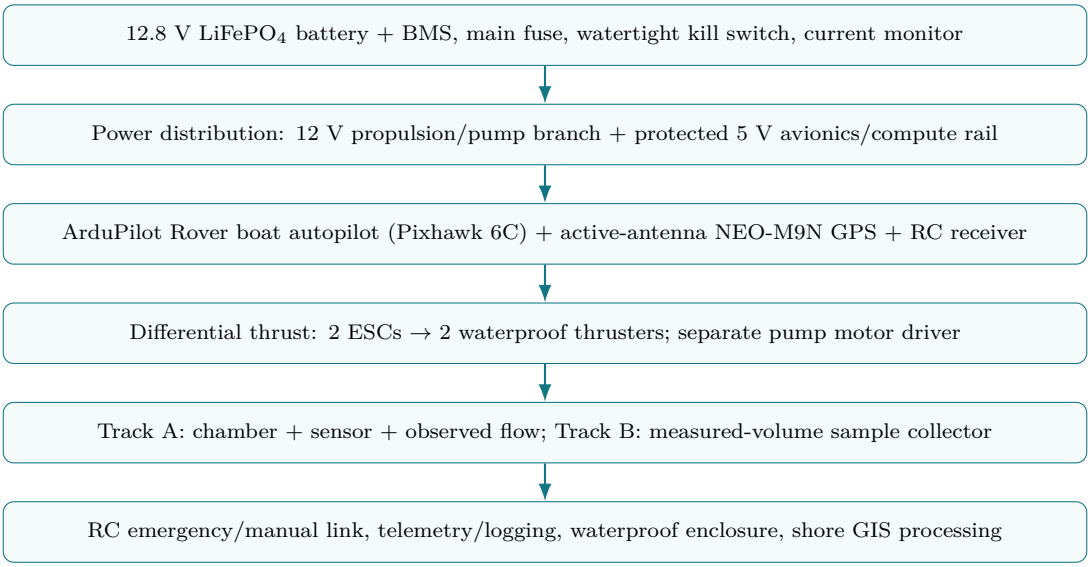


Figure 1: Block diagram (logical, not complete wiring schematic). RC safety and boat-side logs must not depend on the data-telemetry link.

Table 3: Prototype system purchase priorities; masses are budget allowances unless specified by manufacturers.

Critical system	Candidate hardware	Procurement/check before commitment
Hull/buoyancy	2 sealed cylindrical pontoons, 0.90 m long ×0.20 m OD, rigid platform	Measure true sealed displacement, frame strength and wet load
Propulsion	2 Blue Robotics T200 + bidirectional ESCs	12 V compatible, guards, differential steering and sufficient BMS peak current **T200**
Autopilot	Holybro Pixhawk 6C, ArduPilot Rover firmware	Supported hardware; boat mode uses `FRAME_CLASS=2` **Pixhawk6C**, **ArduBoat**
RC link	RadioMaster Pocket 2.4 GHz ELRS with matched serial receiver	Manually test kill/hold/failsafe; ensure regional radio compliance **RadioMaster**
Telemetry	2.4 GHz Wi-Fi for near-bank test; optional compatible SiK pair	MAVLink/logs and onboard SD backup, independent of RC **SiK**
GNSS	u-blox **NEO-M9N-00B** module + compatible powered active L1 antenna/compass unit	Verify receiver board has antenna bias circuitry, magnetometer and correct autopilot cable **M9N**, **SparkM9N**
Sampling	SEAFLO 21-Series 12 V diaphragm pump + calibrated flowmeter	1.2 GPM nameplate is open flow; flow through filters may be lower **Seaflo21**
Power	12.8 V 20 Ah $LiFePO_4$, integrated BMS, $LiFePO_4$ charger	Peak current and fuse coordination; avoid powering motor from flight-controller rail **Bioenno**, **PM02D**
Safety	Sealed electronics, buoyant tether/recovery aid, prop guards, water-leak detector	Safety switch, loss-of-RC configuration, emergency hand retrieval

4 Price-backed critical-component comparison (approved suppliers only)

4.1 MP proxy measurement, water intake and flow verification

Table 4: Monitoring and sampling candidates. Prices are per unit unless stated; every row includes an approved-vendor product hyperlink.

Part and purpose	Supplier / link	Price RM	Selection	Engineering comparison and test gate
DFRobot SEN0189 analogue turbidity	Cytron / listing	55.00	Bench test	Affordable optical turbidity *proxy*. Must calibrate against NTU and laboratory MP counts; not a microplastic-specific detector. Probe sealing and optical fouling need verification.
Seeed S-DTS210-01 industrial RS485 turbidity	Cytron / listing	1,135.00	Field option	IP68, 12–24 V, Modbus/RS485, 0–1000 NTU; stronger environmental packaging but still turbidity rather than chemical MP identification. Price from Cytron Malaysia search page; specifications are shown separately by the same supplier's regional store.
R385 12 V diaphragm pump, 1.6 L/min rated	Cytron / listing	10.90	Bench test	Low purchase cost, 6 W stated; 1.6 L/min is not validated flow through the sampling circuit. Check intake lift, tubing compatibility and contamination blanks.
Micro submersible pump, 3–5 V	Cytron / listing	3.90	Reject field	1.2–1.6 L/min advertised; must be submerged, supplied tube is described as non-food-grade, and suitability for environmental particle sampling is unknown.

Part and purpose	Supplier / link	Price RM	Selection	Engineering comparison and test gate
24 V peristaltic pump, $>$ 90 mL/min	Cytron / listing	49.00	Precision only	Isolates fluid path to tube, but nominal speed is inadequate for fast one-litre stations; requires 24 V or a suitable converter.
YF-S201 pulse flow sensor, 1–30 L/min	Cytron / listing	49.00	Conditional	Low-cost in-line volume audit; stated lower range 1 L/min means it is *not rated* for the 0.5 L/min Track A model. Verify head loss, calibration and microplastic retention.

Decision: Buy the **RM55 turbidity module and RM10.90 R385 pump for bench comparisons**, with a graduated-cylinder calibration; consider the RM1,135 IP68 sensor only after establishing a predictive relationship to laboratory MP. Do *not* combine the 0.5 L/min Track A assumption with the YF-S201 and claim flow accuracy: its published operating range begins at 1 L/min. For low flows use timed gravimetric volume determination or request a certified low-flow meter via RS/Element14/DigiKey. All plastic wetted parts require field and process blanks.

4.2 Position, navigation, imaging and manual control

Table 5: Controller and sensing comparisons. "RFQ" means no approved-vendor public price was verified.

Component	Supplier / link	Price RM	Decision	Technical fit and limitations
GY-NEO6MV2 / NEO-6M satellite receiver	Cytron / listing	21.50	Budget only	UART GNSS and antenna; older single-system GPS option, no advertised compass. No field proof of sub-metre mapping accuracy.
Waveshare LC76G multi-GNSS (AG3352)	Cytron / listing	69.00	Bench pref.	Multi-constellation receiver; listing states 1.5 m CEP and max 10 Hz. No compass or verified ArduPilot plug-and-play cable; survey accuracy requires field tests.
NodeMCU ESP32 stand-alone maker board	Cytron / listing	29.00	Sensor MCU	Used as an ancillary sensor/data-logging controller; *not* a turnkey waypoint/return-to-home autopilot. Quoted price is for the listed NodeMCU variant; other bundles differ.
Holybro Pixhawk 6C + compatible M9N/compass	Registered suppliers / RFQ; maker specification	RFQ	Boat autopilot	Safety-oriented waypoint and RC override architecture from V1. No confirmed approved-supplier public price; request an itemised local quote with correct GPS, power module and firmware compatibility.
HotRC 6-channel 2.4-GHz TX/RX set	Cytron / listing	109.00	Manual test	Listed for model cars/boats, receiver included, PWM output. Confirm *actual* Pixhawk input compatibility (e.g. receiver serial protocols or PPM converter), signal loss failsafe and region compliance.
RadioMaster Pocket ELRS + compatible receiver	Registered suppliers / RFQ; maker product	RFQ	Alternative	V1 reference platform; must quote both handset and matching receiver through registered supplier, not order directly from unlisted manufacturer.
ESP32-CAM with USB connectivity	Cytron / listing	45.00	Bench camera	Promotional web price observed; USB-enabled variant, but limited optics and onboard compute. Recheck at ordering.

Component	Supplier / link	Price RM	Decision	Technical fit and limitations
ESP32-CAM with OV2640 (module only)	GI Electronic / listing	37.00	Alternative	GI product page lists RM37.00 and module/camera; its homepage displayed a different headline price, so use the *product-detail price* and reconfirm by quotation. Additional programmer may be required.

Decision: Navigation needs an independent, reliable RC stop/override path. A low-cost ESP32+LC76G can log position and trial telemetry but does not automatically replace Pixhawk's mission and failsafe logic. Surface RGB imagery of brown or turbid water is *not* evidence that microplastics are directly observable in such images; any image–turbidity–MP regression must be built from matched laboratory references, across water-colour and lighting conditions.

4.3 Power, propulsion and water exclusion

Table 6: Power and hull-related candidate comparisons; no unverified marine propulsion price is inserted.

Component	Supplier / link	Price RM	Decision	Engineering fit and limitation
12 V, 7.2 Ah sealed lead-acid	Cytron / listing	59.00	Bench only	Nominal 86.4 Wh; listed mass about 2.324 kg. Poor two-hour margin at 88 W even with ideal capacity; not interchangeable with proposed $LiFePO_4$.
3S LiPo, 11.1 V, 2.2 Ah, 30C	Cytron / listing	75.00	Short test	24.4 Wh nominal. High advertised C-rate does not provide two hours' capacity; no built-in protection per listing.
3S LiPo, 11.1 V, 5.2 Ah	Cytron / listing	155.00	Short test	57.7 Wh nominal; at 88 W and 80% assumed usable gives about 0.53 h *ideal* endurance, before losses. Requires compatible balance charging and voltage protections.
12.8 V, 20 Ah $LiFePO_4$ + BMS	Registered suppliers / RFQ; reference model	RFQ	Field design	256 Wh nominal per V1 design; request exact BMS current, charger, mass and price. The maker link is a *specification example*, not an authorised purchasing channel.
Balance charger B6AC OEM	Cytron / listing	95.00	Bench option	LiPo/LiFe support shown; verify plug, balance harness and pack-specific charge limits. Must not assume all packs have correct balance taps.
GM25 12 V brushed geared motor	Cytron / listing	49.00	Not a marine thruster	Two would cost RM98 for motors alone; no immersion rating, sealed-shaft design or propeller performance verified. Only consider if designing a protected, shaft-sealed mechanical drive.
MD10C brushed driver, one channel	Cytron / listing	64.26	Brushed only	Two independent units cost RM128.52; suitable *only* for brushed motors within actual current rating, not Blue Robotics three-phase thrusters.
MDD10A brushed driver, two channels	Cytron / listing	108.78	Brushed only	One board controls two brushed motors; each 10 A continuous rated. Ensure motor stall current; *not* a brushless ESC.

Component	Supplier / link	Price RM	Decision	Engineering fit and limitation
Blue Robotics T200 + matching bidirectional ESC	Registered suppliers / RFQ; maker specification	RFQ	V1 field basis	7–20 V marine thruster; maker states 17 A at full throttle 12 V *per thruster*. Request *two* motors, two ESCs, sealed cabling and approved-supplier quote.
IP65 ABS electronics enclosure	Cytron / listing	35–41	Prototype	Price depends on enclosure variant; actual selected configuration and cable glands must be confirmed. IP65 is not submersible.

Decision: retain a *quotation-controlled field drivetrain* instead of selecting low-cost unsealed gearmotors as drop-in substitutes. Never connect a T200 brushless motor to a Cytron MD10C/MDD10A brushed driver. A pair of T200s has approximately 34 A full-throttle current at 12 V before pumps and electronics; V1's illustrative 30 A BMS is undersized for that operating point. Set current limits or revise the battery design after bench measurements.

5 Radio transmission: RC control versus basic telemetry

Table 7: Two distinct links; ranges depend on water-level antenna placement and site conditions.

Function	Example	Design consequence
Manual RC override	RadioMaster Pocket, ExpressLRS 2.4 GHz receiver; CRSF serial input	Priority for emergency stop/hold/manual steering; independently test receiver loss, binding and channel assignment **RadioMaster**
Basic data transmission	Short-range 2.4 GHz Wi-Fi to shore, or paired SiK MAVLink radio	Send time/GPS/battery/QC/concentration updates; do not stream high-resolution raw video unless capacity verified **SiK**
Onboard resilience	SD logging in autopilot or companion computer	Store raw sensor files and logs even when telemetry is lost
Frequency compliance	Malaysia MCMC Class Assignment No. 2 of 2025	Confirm permitted band, module approvals and transmit power before using 433/868/915 MHz **MCMC**

Proposed telemetry packet (not a standard specification): mission ID, timestamp, latitude, longitude, fix quality, battery V/A, motor duty, flow rate, sample ID, counted events, confidence and fault bits. At one compact packet per second, average data rate should be small; verify actual payload and overhead. Record all high-bandwidth images locally.

6 Autopilot versus in-house control board

Table 8: Drone flight controller reuse is supported, but a generic drone ECU is not automatically boat-ready.

Criterion	Pixhawk 6C / ArduPilot Rover	In-house ESP32 / STM32 MCU
Boat software	Official boat variant uses `FRAME_CLASS=2`; Rover supports GPS waypoints, Auto, Guided, RTL and Loiter **ArduBoat**	Must implement navigation, steering loops, state estimation and failsafes
Hardware	STM32H743, multiple PWM, serial/CAN/GPS interfaces; supported ArduPilot releases **Pixhawk6C**, **PixhawkFirmware**	Low-cost and flexible I/O, but architecture is student-developed
Twin-motor drive	Servo-output differential thrust configuration supported with correct vehicle setup	Two ESC PWM signals; differential thrust and mixing coded manually
Mission risk	Mature navigation stack; test firmware, prop guards and mode switching	More coding and test burden, higher risk under two-hour deadline
Recommended division	Safety navigation, waypoint execution and RC override	Sampling scheduler, sensor data, flowmeter and CSV telemetry

The supplied self-driving boat tutorial is a useful build example, but *the governing technical confirmation is ArduPilot's official boat configuration documentation.* Parameter/firmware compatibility must be checked on the chosen board. Set motor outputs, steering and failsafes on the actual boat, not by copying an aircraft mixer.

7 GPS: active antenna, receiver model and mapping implications

Specify the **NEO-M9N-00B** multiconstellation receiver (or Holybro M9N GPS/compass kit with verified module revision). A breakout with *active-antenna bias* is required if using a powered L1 GNSS patch; a bare antenna connector does not guarantee power to an active antenna. u-blox documents support for active antennas, and SparkFun offers NEO-M9N breakouts with external antenna connectors **M9N**, **SparkM9N**. Representative manufacturer positioning accuracy is roughly 1.5 m horizontal CEP under specified conditions **M9N**. This is substantial relative to proposed 5 m grid spacing; repeat ground-point checks, keep GNSS well above the waterline, separate from motors, and log actual fix quality. If sub-metre mapping is essential, upgrade to an RTK-capable *different* receiver and surveyed reference corrections rather than claiming the NEO-M9N provides RTK.

8 Motor comparison: thrust is not hull payload capacity

An electric thruster's static thrust (kgf) balances *horizontal drag*, not the vertical weight that the hull must float. The allowable carried mass is governed primarily by sealed displacement and stability. Therefore the table compares **manufacturer thrust and full-throttle electrical load**; it cannot supply a legitimate "maximum payload kg" without hull geometry, drag and target speed.

Table 9: Manufacturer-rated propulsion comparison. Full-throttle figures must not be interpreted as average cruise consumption.

Motor option	Full forward thrust	Full-load electrical at stated V	Proposed load class*	Implication
2 × T200 (12 V) **T200**	3.71 kgf *each*	17 A, 205 W *each*	Prototype 15–25 kg	Good steering authority, *requires throttle limit and peak-current validation*
2 × T500 (12 V) **T500**	6.0 kgf *each*	16.9 A, 202 W *each*	Heavier hulls, >25 kg	Larger/heavier, likely overkill for sheltered 20-m survey
Minn Kota Endura C2 30 (12 V) **MinnKota**	30 lbf ≈ 13.6 kgf total	up to 30 A, ≈ 360 W	Larger slow platform	Manual tiller/steering adaptation and weight; no differential drive by default

*The "load class" is a proposed *prototype trial mass*, not a rated motor payload or a verified towing capacity. Select actual motors after tow/drag, current and turning tests.

Measured-load test: add 5-kg ballast steps to the completed hull, measure straight-line speed, DC power, steering response, stopping, and recovery. Record operating powers at e.g. 0.15, 0.25 and 0.40 m/s rather than fabricating a vendor speed–payload curve.

9 Hydrostatics and expected payload mass

9.1 Twin-pontoon displacement sizing

For two sealed circular cylinders, $L = 0.90$ m and $D = 0.20$ m:

$$V_{\max} = 2\pi(D/2)^2 L = 0.05655\,\mathrm{m}^3 \approx 56.55\,\mathrm{L}.$$

In fresh water, theoretical maximum fully submerged displaced water mass is approximately 56.55 kg. Using no more than **60% of that displacement as an initial working buoyancy budget** gives about **33.93 kg** supporting total vessel and payload. This is not a hull stability certification: caps, leaks, trim, loading location and waves may reduce available margin.

Table 10: Illustrative all-up wet mass budget (`data/payload_estimate.csv`); validate every line after procurement.

Item	kg	Item	kg
Two 0.9-m pontoons	2.30	Frame and enclosure	2.30
2 T200 + 2 ESCs	1.15	Autopilot/GPS/radios	0.45
Pump, tubing, filters	1.27	Camera and compute	0.40
12.8 V 20 Ah $LiFePO_4$	2.40	Wiring/fuses/extra sensors	0.90
Empty bottles	0.80	Eight litres retained water	8.00
Build contingency	2.00		
Total wet mass	**22.0**	Working buoyancy budget	**33.93**

Important: Track B with *16 litres retained* instead of 8 adds another 8 kg, raising the vessel to about **30.0 kg**. It is still under the 33.93-kg illustrative limit but with poor margin for wave, trim and modelling error. An approach with small bottles, fewer stations or progressive sample unloading is preferable to carrying 16 L simultaneously.

10 Water-pump selection and operating load

Table 11: Published pump specifications. Quoted open-flow rates are not actual filtered flow; record true Q under backpressure, intake lift and fouling.

Model	Supply	Flow rating	Electrical load	Use/risk
Adafruit 1150 peristaltic **Adafruit1150**	12 V	up to 0.10 L/min	0.2–0.3 A = 2.4–3.6 W	Low contamination contact; too slow for many 1-L stations
SEAFLO 21 SFDP1-012-035-21 **Seaflo21**	12 V	1.2 GPM ≈ 4.5 L/min	4.0 A max ≈ 48 W	Candidate collection pump; intermittent duty; measure real flow at filter
SEAFLO 500 GPH bilge SFBP1-G500-01 **SeafloBilge**	12 V	500 GPH nominal	1.6 A nominal; 4 A max	High throughput, low head, not precision metering; needs flowmeter
SEAFLO 33-Series SFWP1-030-070-33 **Seaflo33**	12 V	3.0 GPM ≈ 11.3 L/min	18 A max ≈ 216 W	Heavy duty, high current; unnecessary for initial small-scale sampling

Chosen design basis: SEAFLO 21 for Track B filling or a bypass-controlled Track A stream, with *measured* flow ≥ 1.5 L/min for Track B in the timing example. Sensitive microchannels may need a *separate low-flow metering stage*; do not force unfiltered river water through them. An inline flowmeter plus calibration against graduated water volume is compulsory. Provide coarse guard, cleanable strainers and a properly routed overflow/backflush. Some pump-contact materials are polymers: use field/process blanks and assess sample contamination.

11 Power architecture, management, charging and endurance

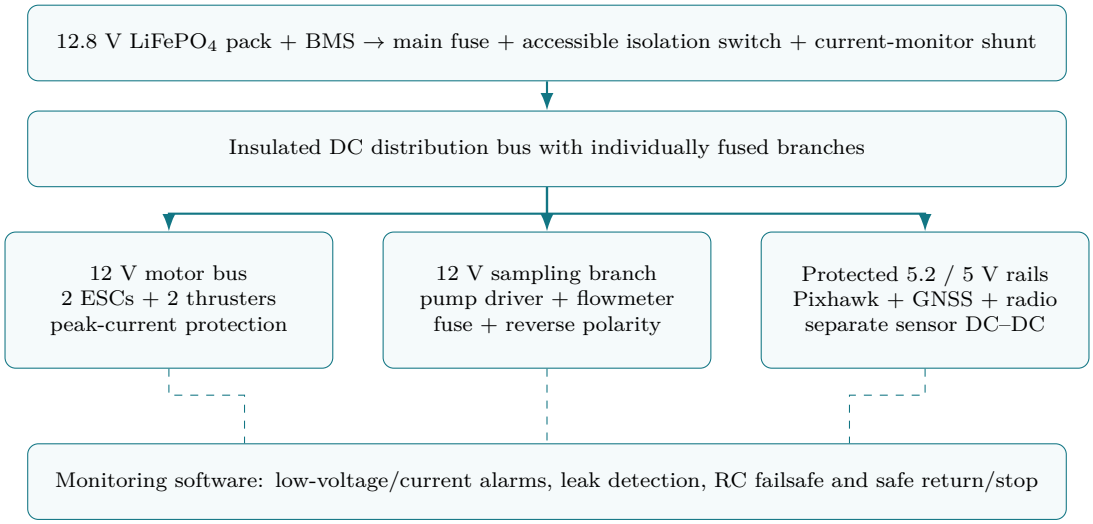


Figure 2: Power-distribution concept with genuinely parallel fused motor, pump and logic branches. Dashed paths denote monitoring/fault signalling rather than physical series wiring.

Controller power: The Holybro PM02D offers regulated 5.2 V up to 3 A from a 2S–12S input, but is *not* a high-current supply for servos, pump or motors **PM02D**. Verify analog/digital module type against the selected Pixhawk connector. Use fused motor and pump circuits; fuses protect wires, not simply the device. **Battery charging:** A representative Bioenno 12 V 20 Ah $LiFePO_4$ pack includes a protection module with 30 A continuous discharge and specifies a suitable 14.6 V charger (manufacturer example 4 A) **Bioenno**. Do not charge $LiFePO_4$ with an unmatched lead-acid or LiPo charger. Charge in a dry supervised location, disconnected from the boat loads, with supplier-approved charger and appropriate temperature constraints.

Table 12: Averaged survey-power allowances at the DC bus: *engineering assumptions*, not manufacturer measurements.

Component / combined function	Track A W	Track B W
Two thrusters at low-speed survey (combined)	50	30
Autopilot, GNSS, RC receiver	6	6
Telemetry / logging	4	2
Camera/computer	9	0
Pump, *duty-averaged*	12	6
Conversion/misc. margin	7	4
Estimated average	**88**	**48**

$$E_{\text{nom}} = 12.8 \times 20 = 256\,\text{Wh}, \qquad E_{\text{usable}} = 0.80 \times 256 = 204.8\,\text{Wh}, \qquad t = E_{\text{usable}}/P_{\text{average}}.$$

Table 13: Illustrative 12.8 V, 20 Ah battery operating time: energy-only estimates, ignoring rate/thermal limits and unmodelled loads. Source: `data/power_endurance.csv`.

Scenario	Avg W	h	Energy for 2h	2h energy gate
Track B economical	48	4.27	96 Wh	Pass
Track A planned	88	2.33	176 Wh	Pass (narrow)
Track A difficult water	125	1.64	250 Wh	Fail
High-drive manoeuvres	180	1.14	360 Wh	Fail
Full-throttle dual T200 at 12 V + loads	430	0.48	860 Wh	Fail; BMS also inadequate

Critical current warning: Two T200 thrusters at maximum 12 V load could draw approximately $2 \times 17 = 34$ A before other components. The representative Bioenno 20 Ah pack is rated 30 A continuous; full power on both

thrusters is therefore **not permissible** with this battery/BMS. Throttle-limit the ESCs or select a pack with a verified higher discharge rating, suitably rated wires and protection. The 88 W Track A estimate requires validation with an inline wattmeter on the water, including startup and snagged-propeller cases. **T200**, **Bioenno**

12 Time budget and survey choices

Track A illustrative path: five 20-m lanes plus four 5-m lateral moves gives 120 m. At an assumed 0.25 m/s the ideal transit is 8 min. A 60-min effective sensing period plus 25 min setup, turns, recovery, QC and map generation totals 93 min. A 0.50-L/min fully interrogated flow yields 30 L in the sensing period, *not* 30 L per map cell. Sensor sampling time and intake-delay calibration matter.

Track B illustrative workflow: 16 stations, 1 L per site, effective pumping 1.5 L/min, 1.2 min/station for handling and 3 min/station portable image processing; add 18 min survey movement/setup/mapping. The total is about 95.9 min. Changing image time to 5 min/site raises it above 120 min. Field concentration and sampling statistics impose a stricter constraint: at around 3.12 particles/L, a 1-L sample expects only 3.12 particles, equivalent to 57% approximate Poisson standard error *before* sensor losses. The first demonstration should specify a minimum reporting volume and show uncertain/insufficient-count cells rather than inventing a smooth precise heat map.

13 Risk controls, procurement order and decision

1. **Float test first:** assemble sealed pontoons, weigh all components, add ballast to 22 and 30 kg, check reserve freeboard, trim and wave splash under stationary and moving load.
2. **Then drive/power test:** two T200 and ESCs (or lower-cost alternative only with measured equivalent performance), current limit and speed tests at full mass. Inspect for overheating and waterproof cable glands.
3. **Autopilot and RC:** flash appropriate Rover firmware, configure boat class and motor layout, test manual/RTL/Loiter and safe response to link or GNSS loss while tethered.
4. **Sample circuit:** bench-test the selected pump's *actual* flow through chosen strainers/filter and tubing at target depth; verify sampled volume and cleaning blanks.
5. **Timing and credibility:** time full 16-sample shoreline chain or Track A effective observed-volume throughput; reject any concept unable to create an auditable map within 120 min.
6. **Procurement and local radio:** request quotations from university-registered electronics/hardware suppliers, confirm Malaysia radio approval, battery transport/charging and replacement parts.

V1 engineering reference architecture, pending V2 approved-supplier quotations: two 0.9-m × 0.20-m sealed pontoons, two T200 thrusters with compatible reversible ESCs, Pixhawk 6C running ArduPilot Rover boat firmware, external-active-antenna NEO-M9N (GPS/compass integration), ELRS 2.4-GHz RC control, separate low-bandwidth telemetry and local logs, SEAFLO 21 pump plus flowmeter, and 12.8 V 20 Ah $LiFePO_4$ with correct charger and managed current limits. *No full 16-L sample collection without revising the wet-mass, freeboard and storage plan.* The expected 22-kg wet all-up budget is an engineering starting value, not an independently confirmed built boat.

14 V2 procurement-ready basket and transparent cost gap

The prices below are *a purchase-planning example*, not a full field-boat quotation. This basket supports **bench calibration, low-power position logging and a short-duration manually controlled proof-of-concept**. The selection uses one reasonably priced component from each function; it deliberately does not claim that a short-test LiPo is capable of a 120-minute field mission.

Table 14: One-unit price-backed bench/prototype basket. Each hyperlink leads to the approved supplier's specific item listing. All values in RM (2 November 2026).

Component (supplier hyperlink)	Qty	Unit RM	Line RM
NodeMCU ESP32, Cytron	1	29.00	29.00
LC76G multi-GNSS, Cytron	1	69.00	69.00
SEN0189 turbidity, Cytron	1	55.00	55.00
R385 pump, Cytron	1	10.90	10.90
YF-S201 flowmeter, Cytron	1	49.00	49.00
ESP32-CAM, GI Electronic	1	37.00	37.00
IP65 box (small variant), Cytron	1	35.00	35.00
HotRC TX + RX, Cytron	1	109.00	109.00
3S 5.2 Ah LiPo, Cytron	1	155.00	155.00
B6AC balance charger, Cytron	1	95.00	95.00
Documented listed-price subtotal			**643.90**

Cost exclusions: delivery/tax, water-safe hoses and fittings, filter and reference-counting consumables, buffer and calibration chemicals, reference microscopy, proper waterproof cable glands, sensor ADC if needed, fuse/wire/kill switch, motor/thruster mounts, buoyant hull and frame, battery protection, sensors for leak/current, waterproof electronics mounting, field-certified power supply and, above all, **2 marine thrusters + 2 matching brushless ESCs + Pixhawk 6C / power module**. The RM643.90 is *not* a full floating survey system price, and the YF-S201 would not validate the 0.5 L/min flow assumption.

14.1 Three staged purchase decisions

1. **Purchase now only after approval (low-risk):** turbidity module, R385, ESP32, GNSS, calibrated beaker/scale; use indoor tap-water and microplastic reference trials before installing the components on a boat.
2. **Quote before hull freeze (high-risk):** 2 T200-compatible thrusters/ESCs *or* a fully specified waterproof alternative; Pixhawk/GNSS/compass assembly, RC interface, 12.8 V 20 Ah $LiFePO_4$ with current-appropriate BMS/charger; two sealed pontoons and waterproof fittings. Solicit through Cytron, Element14, RS, DigiKey, Autobotic, Mybotic, Vanguard and the registered hardware contacts; accept only an actual supplier response as a price.
3. **Defer until calibration evidence:** RM1,135 industrial optical turbidity probe, more expensive image acquisition, GPS accuracy upgrades and retained 16-L sample storage. These require proof that the measured variable predicts actual MP concentrations better than the cheaper prototype.

14.2 RFQ form and verification before raising a purchase order

Table 15: Fields to request from registered suppliers; a missing quotation is a procurement blocker, not a zero-cost line item.

RFQ item	Required written evidence
Marine propulsion set	Exact part model, qty 2 thrust units + qty 2 reversible ESCs, voltage, continuous/start current, guards, cable glands and water rating, unit/total MYR, warranty and lead time
Navigation/RC	Pixhawk 6C revision, GNSS/compass and antenna options, PM02D or correct power module, complete RC receiver data interface and failsafe, availability and purchase total
Energy subsystem	12.8 V 20 Ah $LiFePO_4$ pack, continuous/peak BMS rating, included cell protection, charger model, connector, weight, shipping limitations and total MYR
Hull + wetted parts	Watertight pontoon material, sealed displacements, frame/bulkhead fabrication, fittings, tubing, cleanable mesh, microplastic contamination controls and total MYR

Order-release gate: sign off the selected architecture, full all-up mass, two-hour energy feasibility, wet-sensor QC protocol, safe recovery and current limits *before* spending on the expensive field package. Confirm any import freight and Malaysian radio compliance. Only if no registered supplier can fulfil should the team contact Procurement for a non-registered purchase exception.

15 V2 measurement validity and field-test gates

The uploaded project brief requires a *two-dimensional MP concentration map with demonstrable links to true MP and an appropriate laboratory reference.* The talk supplied by the supervisors describes direct measurements and image-based approaches as open options, and specifically cautions that image methods require a calibration relationship. Accordingly:

1. **Reference protocol:** obtain duplicate, known-volume samples across clean and contaminated water conditions, filter and microscopy-count, and document blanks and recovery. Record concentration as $C_{\mathrm{MP}} = N_{\mathrm{confirmed}}/V_{\mathrm{sampled}}$, with the reporting size window and sample volume.
2. **Proxy protocol:** record contemporaneous turbidity/optical intensity and environmental confounders at the same sample locations and times. Test between-day and across-location generalisation instead of relying on a high in-sample correlation.
3. **Survey protocol:** log fix quality, latitude, longitude, timestamps, transect ID, stage of processing, flow/volume and uncertainties; map blank/insufficient-data cells explicitly instead of interpolating false precision.
4. **Time and payload:** preserve the V1 two-hour and $20 \times 20\,\mathrm{m}$ target, but qualify any result as a measured count, calibrated estimate or *uncalibrated turbidity proxy*; they are not interchangeable.

Critical V2 change relative to V1: low-cost products are explicitly *calibration and development candidates*, not automatic replacement for marine-rated navigation, propulsion, power or an MP-specific laboratory assay. All compared item prices are linked to an approved supplier, and the three high-value unresolved packages remain labelled RFQ.